

Project Proposal: Statistical Inference on Medical Results using the Framingham Heart Study Dataset

Statistical Inference Project
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1 Introduction

Cardiovascular disease (CVD) remains a leading cause of morbidity and mortality globally. This project proposes a comprehensive statistical inference analysis utilizing the celebrated Framingham Heart Study dataset. The primary objective is to apply rigorous statistical methods—ranging from hypothesis testing to regression modeling—to understand the risk factors associated with coronary heart disease. By examining the relationships between demographic, behavioral, and medical variables, this study aims to derive generalized conclusions about cardiovascular health in the population.

2 Data Provenance and Structure

2.1 Origin and Source

The dataset selected for this analysis originates from the **Framingham Heart Study**, a long-term, ongoing cardiovascular study on residents of the town of Framingham, Massachusetts. The dataset was obtained from Kaggle¹.

2.2 Goal and Methodology

The original goal of the study was to identify common factors or characteristics that contribute to cardiovascular disease (CVD) by following its development over a long period of time.

The study design is a **longitudinal cohort study**. While this methodology is powerful for establishing temporal sequence and inferring causality, it is subject to potential biases. Most notably, *selection bias* is inherent due to the geographic restriction to a specific town in Massachusetts, which may limit the generalizability of findings to broader, more diverse populations. Furthermore, attrition bias may occur if subjects drop out of the study over time.

2.3 Data Dictionary

The dataset comprises approximately 4,240 observations and 16 variables. The variables are classified as follows:

Categorical Variables (Nominal/Ordinal):

- **male:** Gender of the participant (0 = Female, 1 = Male).
- **education:** Level of education (Ordinal: 1 = Some High School, 2 = High School/GED, 3 = Some College/Vocational School, 4 = College).

¹<https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset/data>

- **currentSmoker**: Smoking status (0 = No, 1 = Yes).
- **BPMeds**: Whether the patient is on blood pressure medication (0 = No, 1 = Yes).
- **prevalentStroke**: History of stroke (0 = No, 1 = Yes).
- **prevalentHyp**: History of hypertension (0 = No, 1 = Yes).
- **diabetes**: Diabetes status (0 = No, 1 = Yes).

Continuous Variables (Ratio/Interval):

- **age**: Age of the patient at the time of examination.
- **cigsPerDay**: Number of cigarettes smoked per day.
- **totChol**: Total cholesterol level (mg/dL).
- **sysBP**: Systolic blood pressure (mmHg).
- **diaBP**: Diastolic blood pressure (mmHg).
- **BMI**: Body Mass Index (kg/m^2).
- **heartRate**: Heart rate (beats per minute).
- **glucose**: Blood glucose level (mg/dL).

Target Variable:

- **TenYearCHD**: The 10-year risk of coronary heart disease (Binary: 0 = No, 1 = Yes).

3 Suitability for Statistical Inference

3.1 Generalization

This dataset is highly suitable for statistical inference due to its substantial sample size ($n \approx 4,240$). According to the Central Limit Theorem, this size allows for robust statistical power and ensures that the sampling distribution of the sample means approximates normality, facilitating the use of parametric tests. The dataset's granularity allows for the detection of subtle effect sizes that might be missed in smaller cohorts.

3.2 Assumptions and Limitations

To perform valid inference, standard assumptions must be checked. We assume **independence of observations**, implying that one patient's health outcomes do not influence another's.

However, we must address potential violations of normality. Variables such as **glucose** and **BMI** often exhibit right-skewed distributions in medical data. We will assess normality using visual inspection (Q-Q plots) and statistical tests (Shapiro-Wilk). If normality is violated, we will employ data transformation techniques (e.g., Log-transformation) or utilize non-parametric alternatives (e.g., Mann-Whitney U test) to ensure the validity of our conclusions.

4 Scope of Analysis

This project will execute five distinct statistical inference analyses to investigate different facets of cardiovascular health.

4.1 Hypothesis Testing: Blood Pressure and Smoking

Objective: To determine if there is a statistically significant difference in Systolic Blood Pressure (**sysBP**) between smokers and non-smokers. **Method:** Independent Samples T-test (or Mann-Whitney U test if normality is violated). **Justification:** Smoking is a known vasoconstrictor. Testing this hypothesis validates the dataset against established medical knowledge and quantifies the physiological impact of smoking on blood pressure within this specific cohort.

4.2 Analysis of Variance (ANOVA): Education and Obesity

Objective: To compare the mean Body Mass Index (BMI) across the four levels of `education`. **Method:** One-way ANOVA followed by Tukey's HSD post-hoc test if significant differences are found. **Justification:** Socioeconomic status, often proxied by education, is frequently linked to health outcomes. This analysis investigates whether higher educational attainment correlates with lower BMI, potentially indicating better access to health resources or nutritional knowledge.

4.3 Logistic Regression: Predicting Coronary Heart Disease

Objective: To model the probability of a patient developing Coronary Heart Disease (`TenYearCHD`) based on multiple predictors. **Method:** Binary Logistic Regression using `age`, `male`, `sysBP`, and `totChol` as independent variables. **Justification:** Since the outcome variable is binary, linear regression is inappropriate. Logistic regression allows us to estimate the odds ratios for each risk factor, controlling for confounders, and identifying which physiological markers are the strongest predictors of CHD risk.

4.4 Chi-Square Test of Independence: Comorbidities

Objective: To test for an association between Diabetes (`diabetes`) and Hypertension (`prevalentHyp`). **Method:** Pearson's Chi-Square Test of Independence. **Justification:** Both diabetes and hypertension are components of metabolic syndrome. Determining if these conditions are independent or associated is crucial for understanding comorbidity risks. A significant result would suggest that patients with one condition should be rigorously screened for the other.

4.5 Estimation: Population Glucose Levels

Objective: To estimate the true population mean of blood glucose levels. **Method:** Construction of a 95% Confidence Interval for the mean of `glucose`. **Justification:** Point estimates alone can be misleading. A confidence interval provides a range of plausible values for the population parameter, offering a measure of precision. This establishes a baseline for "normal" glucose levels within the demographic represented by the study.

5 Conclusion

This proposal outlines a structured approach to analyzing the Framingham Heart Study dataset. By rigorously applying these five statistical methods, the project will not only confirm known medical associations but also demonstrate the practical application of statistical inference in epidemiology.